Homework 5 (Bonus)

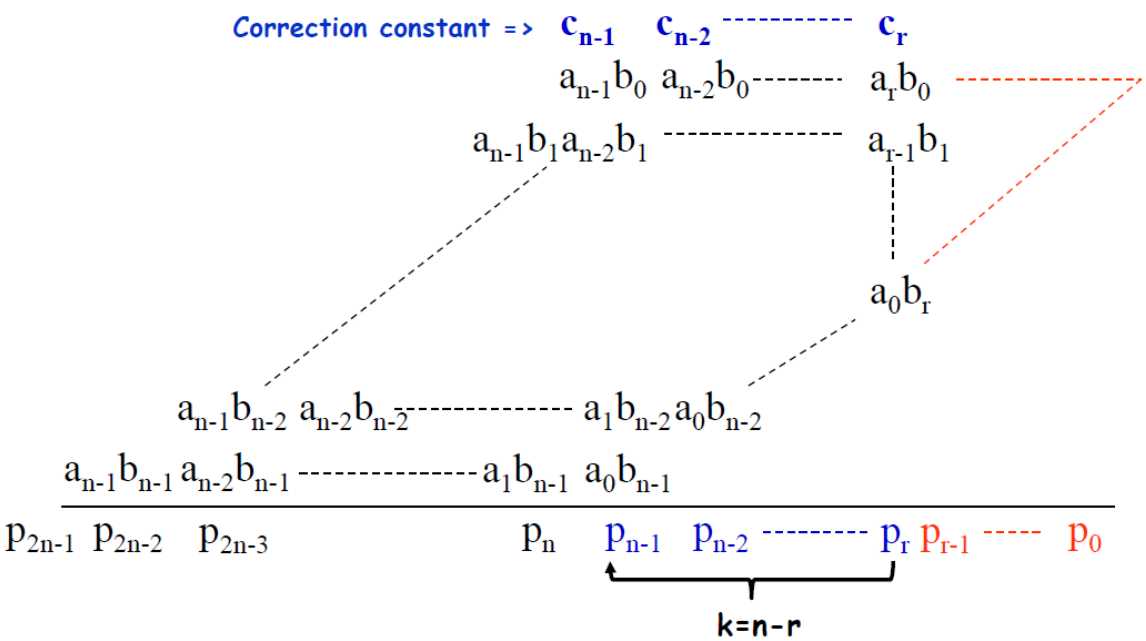
Truncated Multiplier Designs

Handout: 2025/11/24

Due: 2025/12/29

Make comparison for different truncated multipliers for 8x8 and 16x16 for unsigned numbers. Select the value of k and r so that the maximum absolute error is within 1 ulp = 2^{n-1} in case of truncated multiplier designs generating the n-bit most significant half of the product.

(1) Verilog operator * (operator)
(2) Partial product row accumulation using for-loop (row)
(3) Constant-correction truncated multiplier using partial product row accumulation (trunc_const_row)
(4) Array multiplier (array)
(5) Variable correction trunvated array multiplier (trunc_var_array)



- $r = n-k$ least significant columns are eliminated, a correction constant is added, and the product is truncated to n bits
- $k=n-r$ is the number of reserved columns for PP compression

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Fig. 1. Truncated multiplier with constant correction.

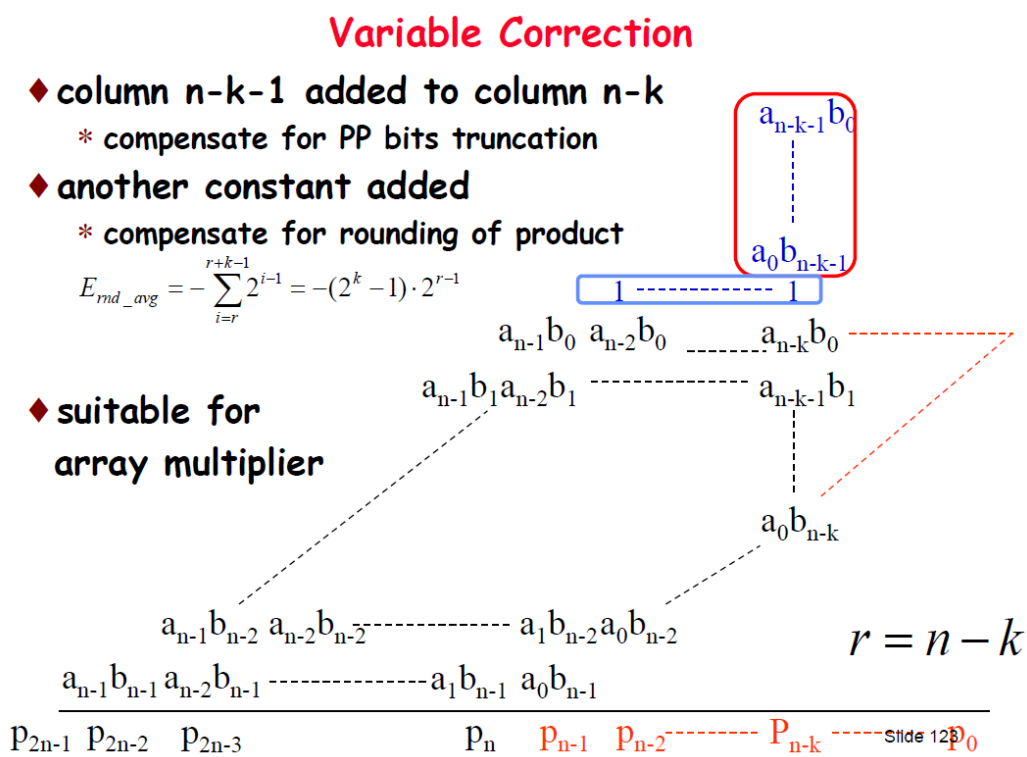
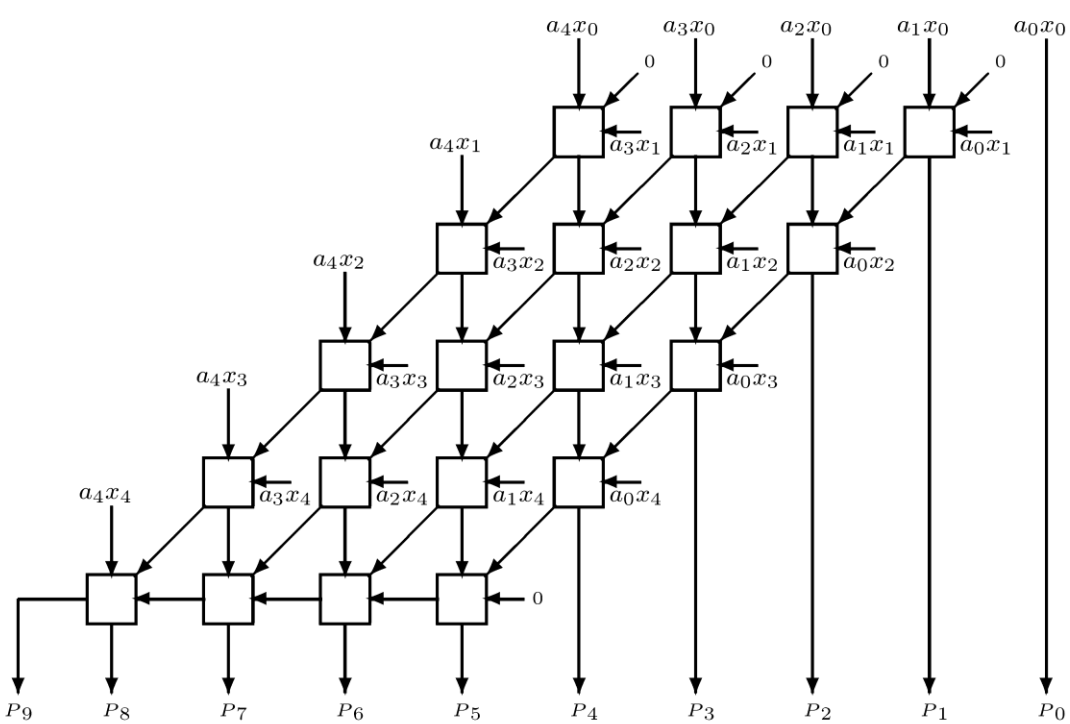
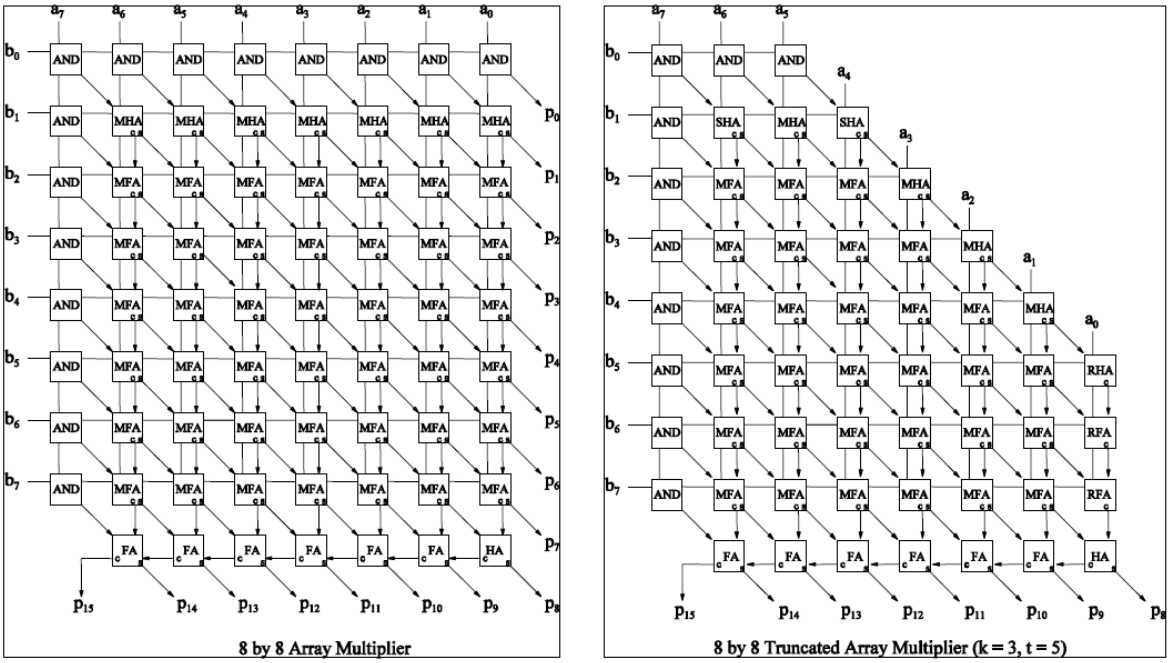


Fig. 2. Truncated multiplier with variable correction.



M. J. Schule, J. E. Stine, and J. G. Jansen, "Reduced power dissipation through truncated multiplication," in IEEE Alessandro Volta Memorial International Workshop on Power Design, pp. 61-69, Mar. 1999.

Truncated Array Multipliers (variable-correction)



MFA (modified FA): AND+FA RFA (reduced FA): FA producing carry out only

MHA (modified HA): AND+HA SHA (specialized HA): MFA with one input set to 1

Fig. 3. (a) 5x5 unsigned array multiplier; (b)truncated multiplier with variable correction and array.

k	Constant Correction	Variable Correction
0	n = 1	1 <= n <= 3
1	2 <= n <= 3	4 <= n <= 7
2	4 <= n <= 6	8 <= n <= 14
3	7 <= n <= 11	15 <= n <= 27
4	12 <= n <= 20	28 <= n <= 52
5	21 <= n <= 37	53 <= n <= 101
6	38 <= n <= 70	102 <= n <= 198

Fig. 4. Selection of k to achieve 1 ulp = 2^n-1.

Fill in the following summarized comparison table for nxn= 8x8 and 16x16

1. nxn unsigned		area(um2)	time(ns)	power(uW)		
		CL(total)	delay	dynamic	leakage	total
operator	area					
	mid					
	delay					
row	area					
	mid					
	delay					
trunc_const_row	area					
	mid					
	delay					
array	area					
	mid					
	delay					
trunc_var_array	area					
	mid					
	delay					